RESEARCH PRESENTATION

Credit Card Fraud Detection Using DBSCAN-Augmented Machine Learning

Based on

Ghalwash et al. (2025).
Enhancing credit card fraud detection using DBSCAN-augmented disjunctive voting ensemble. Scientific Reports, 15, 39754.

Data: Kaggle Credit Card Fraud Detection Dataset 2023

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Credit Card Fraud Detection Using DBSCAN-Augmented Machine Learning

Can clustering-derived features help supervised classifiers detect fraud more effectively?

Imbalanced classification

Fraud is the rare class. Accuracy is uninformative: always predicting legitimate scores 99.01% on the working subset.

Missed fraud vs false alarms

False negatives create financial loss. False positives inconvenience legitimate customers. Both matter.

Question, not a claim

DBSCAN is not assumed to mark fraud as noise. That is an empirical hypothesis tested on a held-out fold.

Dataset: Kaggle Credit Card Fraud Detection 2023

Original file

568,630 rows · 50/50 Class · 0 missing · 0 duplicate rows

Working experiment (paper D2 slice)

80,000 legitimate + 800 fraudulent = 80,800 rows

1% fraud by construction — not the file's natural prevalence

Stratified 80/20 split, seed 42: train 64,640 (640 fraud) · test 16,160 (160 fraud)

Decision trade-off (schematic)

Legitimate

Class = 0

False alarm = customer friction

Fraud

Class = 1

Missed fraud = financial loss

Using DBSCAN to Enrich Transaction Features

Same classifiers, same held-out test fold, three input representations

1 Original features

Train classifiers on V1–V28 and Amount only.

2 Fit DBSCAN on train

Dense regions vs noise (label −1). eps=2, min_samples=4.

3 Two encodings

Numeric cluster IDs (paper-style) or a binary is_noise flag.

4 Paired comparison

Evaluate original vs augmented models on the same test set.

A Original

V1 … V28, Amount

No cluster-derived column.
Baseline for every paired test.

B Paper-style IDs

Original + integer cluster IDs

Keeps cluster identity, but treats arbitrary categorical labels as numbers.

C Leakage-safe flag

Original + binary is_noise

Marks unusual points without giving numeric meaning to cluster IDs.

Hypothesis, not a guarantee

On train, 64.2% of fraud is marked noise — but only 2.0% of noise is fraud (P(fraud | noise) = 0.020). Most noise is still legitimate. DBSCAN is not guaranteed to isolate fraud. Whether the extra column helps is an empirical result.

Experimental Architecture and Model Selection

Single stratified holdout · no cross-validation · no test-set threshold tuning

Validate schema → Build 1% D2 subset → Stratified 80/20 split → Fit scaler on train → DBSCAN features → Train classifiers → Evaluate untouched test

Branch A · paper-style numeric IDs

Fit DBSCAN on scaled train features; append integer cluster IDs; iterate once more.

Out-of-sample: nearest training point's label; if distance > eps, assign noise.

Paper trio: RF n=3, KNN k=3, RBF-SVM C=1. Majority (≥2) and OR (≥1) votes.

Branch B · leakage-safe noise flag

One DBSCAN on train. Feature is binary is_noise (label == −1). No integer IDs.

Same 1-NN + eps gate. Unsupported test points stay unassigned/noise (25.3% of test).

Paired original vs original+flag for LogReg, RF-200, XGBoost, CatBoost, and the paper trio.

Why this model set

Baselines

Logistic Regression: simple linear floor.

RF n=200: unrestricted tree baseline.

XGBoost / CatBoost: modern boosting.

Paper models

RF n=3, KNN k=3, RBF-SVM as specified.

Majority: fraud if ≥2 agree.

OR vote: fraud if any of the three fires.

Protocol

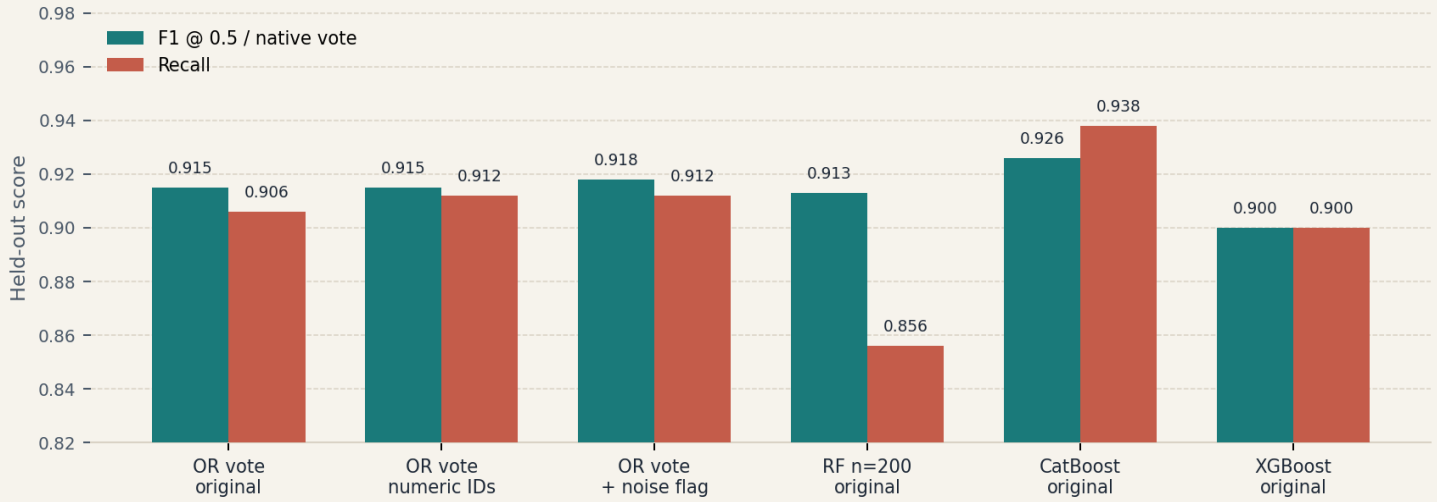
Scaler, DBSCAN, and models fit on train only.

Trees use raw V*/Amount ± extra column.

Threshold 0.5 / native predict; votes have no PR-AUC.

Results: Does DBSCAN Augmentation Help?

One held-out test fold · 16,160 rows, 160 fraud · PR-AUC ranks scored models; votes have no PR-AUC



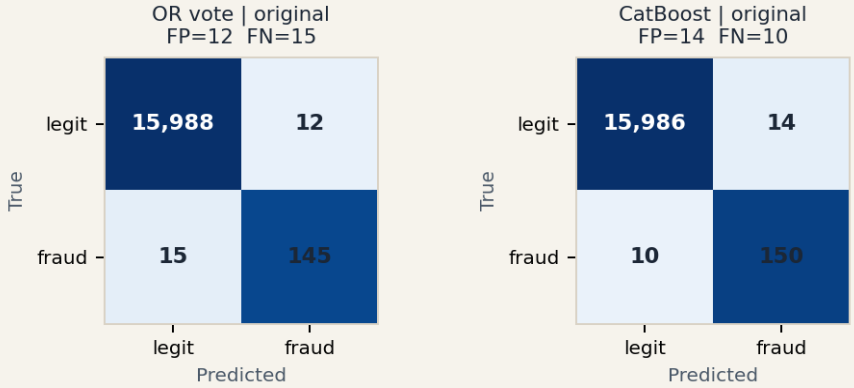
What the paired tests show

Numeric IDs do not lift the OR ensemble (F1 0.915 both) and hurt SVM (F1 0.918 → 0.862).

Noise flag is mixed: OR F1 0.915 → 0.918; CatBoost F1 0.926 → 0.914; RF-200 PR-AUC stays 0.959.

Best scored F1: CatBoost original (0.926). Best non-paper PR-AUC: RF n=200 (0.959).

Model	P	R	F1	PR-AUC	MCC	FP	FN
OR vote | original	0.924	0.906	0.915	—	0.914	12	15
OR vote | numeric IDs	0.918	0.912	0.915	—	0.915	13	14
OR vote | + noise flag	0.924	0.912	0.918	—	0.917	12	14
RF n=200 | original	0.979	0.856	0.913	0.959	0.915	**3**	23
CatBoost | original	0.915	**0.938**	**0.926**	0.958	**0.925**	14	10
XGBoost | original	0.900	0.900	0.900	0.955	0.899	16	16



Trade-off. CatBoost catches more fraud (FN 10 vs 15) at similar false alarms (14 vs 12). RF n=200 is the opposite: only 3 FP, but 23 missed fraud. Accuracy (~0.998) does not distinguish these models.

Conclusions and Next Steps

Findings are for this constructed 1% slice and one seed-42 holdout

DBSCAN did not reliably improve detection

Paired tests are mixed. The noise flag never produced a large PR-AUC gain. Numeric IDs left OR F1 unchanged and damaged SVM.

Noise flag was more stable than numeric IDs

IDs are arbitrary numbers; SVM F1 fell from 0.918 to 0.862. The flag's effects were small, not a consistent lift.

Paper OR is competitive, not dominant

OR + flag F1 0.918 vs CatBoost original 0.926. Majority + flag F1 0.928 with 4 FP, but recall only 0.887.

Operating point, not just a score

RF n=200: 3 FP / 23 FN. CatBoost: 14 FP / 10 FN. Choose by missed-fraud vs alert cost, not accuracy.

Limitations

1% rate is sampled from a 50/50 file.

One stratified split; seed variance unknown.

Paper IDs: representation replica, not exact code match.

OOS assignment is a 1-NN + eps approximation.

Not a deployment or cost-sensitive evaluation.

Future work

Multiple splits or cross-validation.

eps / min_samples and OOS assignment variants.

Validation-based thresholds; calibration.

Natural prevalence data; explicit FP/FN costs.

Takeaway

On this constructed 1% holdout, a well-configured CatBoost or Random Forest on original features matches or beats the paper's DBSCAN-augmented OR ensemble. The extra cluster column is not the main source of performance.